

UNIT-11 Computer Security

- ❑ Computer Security,
 - ❑ Firewall,
 - ❑ Cryptography
- ❑ Cyber Law, Digital
 - ❑ Signature,
- ❑ Certificate Authority
- ❑ Security Awareness, Security Policy

What is computer security?

- Computer security basically is the protection of computer systems and information from harm, theft, and unauthorized use.
- It is the process of preventing and detecting unauthorized use of your computer system.
- Computer security refers to protecting and securing computers and their related data, networks, software, hardware from unauthorized access, misuse, theft, information loss, and other security issues.

Types of computer security

- *Computer security can be classified into four types:*
- **1. Cyber Security:** Cyber security means securing our computers, electronic devices, networks , programs, systems from cyber attacks. Cyber attacks are those attacks that happen when our system is connected to the Internet.
- **2. Information Security:** Information security means protecting our system's information from theft, illegal use and piracy from unauthorized use. Information security has mainly three objectives: confidentiality, integrity, and availability of information.
- **3. Application Security:** Application security means securing our applications and data so that they don't get hacked and also the databases of the applications remain safe and private to the owner itself so that user's data remains confidential.
- **4. Network Security:** Network security means securing a network and protecting the user's information about who is connected through that network. Over the network hackers steal, the packets of data through sniffing and spoofing attacks, man in the middle attack, war driving, etc, and misuse the data for their benefits.

Types of cyber attack

- **1. Denial of service attack or DOS:** A denial of service attack is a kind of cyber attack in which the attackers disrupt the services of the particular network by sending infinite requests and temporary or permanently making the network or machine resources unavailable to the intended audience.
- **2. Backdoor:** In a backdoor attack, malware, trojan horse or virus gets installed in our system and start affecting it's security along with the main file. Consider an example: suppose you are installing free software from a certain website on the Internet. Now, unknowingly, along with this software, a malicious file also gets installed, and as soon as you execute the installed software that file's malware gets affected and starts affecting your computer security. This is known as a backdoor.
- **3.Eavesdropping:** Eavesdropping refers to secretly listening to someone's talk without their permission or knowledge. Attackers try to steal, manipulate, modify, hack information or systems by passively listening to network communication, knowing passwords etc. A physical example would be, suppose if you are talking to another person of your organization and if a third person listens to your private talks then he/ she is said to eavesdrop on your conversation. Similarly, your conversation on the internet maybe eavesdropped by attackers listening to your private conversation by connecting to your network if it is insecure.
- **4. Phishing:** Phishing is pronounced as “fishing” and working functioning is also similar. While fishing, we catch fish by luring them with bait. Similarly, in phishing, a user is tricked by the attacker who gains the trust of the user or acts as if he is a genuine person and then steals the information by ditching. Not only attackers but some certain websites that seem to be genuine, but actually they are fraud sites. These sites trick the users and they end up giving their personal information such as login details or bank details or card number etc. Phishing is of many types: Voice phishing, text phishing etc.

Types of cyber attack Cont..

- **5. Spoofing:** Spoofing is the act of masquerading as a valid entity through falsification of data(such as an IP address or username), in order to gain access to information or resources that one is otherwise unauthorized to obtain. Spoofing is of several types- email spoofing, IP address spoofing, MAC spoofing , biometric spoofing etc.
- **6. Malware:** Malware is made up of two terms: Malicious + Software = Malware. Malware intrudes into the system and is designed to damage our computers. Different types of malware are adware, spyware, ransomware, Trojan horse, etc.
- **7. Social engineering:** Social engineering attack involves manipulating users psychologically and extracting confidential or sensitive data from them by gaining their trust. The attacker generally exploits the trust of people or users by relying on their cognitive basis.
- **8. Polymorphic Attacks:** Poly means “many” and morph means “form”, polymorphic attacks are those in which attacker adopts multiple forms and changes them so that they are not recognized easily. These kinds of attacks are difficult to detect due to their changing forms.

Steps to ensure computer security

- ***In order to protect our system from the above-mentioned attacks, users should take certain steps to ensure system security:***
- 1. Always keep your Operating System up to date. Keeping it up to date reduces the risk of their getting attacked by malware, viruses, etc.
- 2. Always use a secure network connection. One should always connect to a secure network. Public wi-fi's and unsecured networks should be avoided as they are at risk of being attacked by the attacker.
- 3. Always install an Antivirus and keep it up to date. An antivirus is software that scans your PC against viruses and isolates the infected file from other system files so that they don't get affected. Also, we should try to go for paid anti-viruses as they are more secure.
- 4. Enable firewall. A firewall is a system designed to prevent unauthorized access to/from a computer or even to a private network of computers. A firewall can be either in hardware, software or a combination of both.
- 5. Use strong passwords. Always make strong passwords and different passwords for all social media accounts so that they cannot be key logged, brute forced or detected easily using dictionary attacks. A strong password is one that has 16 characters which are a combination of upper case and lower case alphabets, numbers and special characters. Also, keep changing your passwords regularly.
- 6. Don't trust someone easily. You never know someone's intention, so don't trust someone easily and end up giving your personal information to them. You don't know how they are going to use your information.

Steps to ensure computer security

- 7. Keep your personal information hidden. Don't post all your personal information on social media. You never know who is spying on you. As in the real world, we try to avoid talking to strangers and sharing anything with them. Similarly, social media also have people whom you don't know and if you share all your information on it you may end up troubling yourself.
- 8. Don't download attachments that come along with e-mails unless and until you know that e-mail is from a genuine source. Mostly, these attachments contain malware which, upon execution infect or harms your system.
- 9. Don't purchase things online from anywhere. Make sure whenever you are shopping online you are doing so from a well-known website. There are multiple fraud websites that may steal your card information as soon as you checkout and you may get bankrupt by them.
- 10. Learn about computer security and ethics. You should be well aware of the safe computing and ethics of the computing world. Gaining appropriate knowledge is always helpful in reducing cyber-crime.
- 11. If you are attacked, immediately inform the cyber cell so that they may take appropriate action and also protect others from getting attacked by the same person. Don't hesitate to complain just because you think people may make your fun.
- 12. Don't use pirated content. Often, people try to download pirated movies, videos or web series in order to get them for free. These pirated content are at major risk of being infected with viruses, worms, or malware, and when you download them you end up compromising your system security.

Potential Losses due to Security Attacks

Losing you data

- If your computer has been hacked or infected, there is a big chance that all your stored data might be taken by the attacker.

Bad usage of your computer resources –

- This means that your network or computer can go in overload so you cannot access your genuine services or in a worst case scenario, it can be used by the hacker to attack another machine or network.

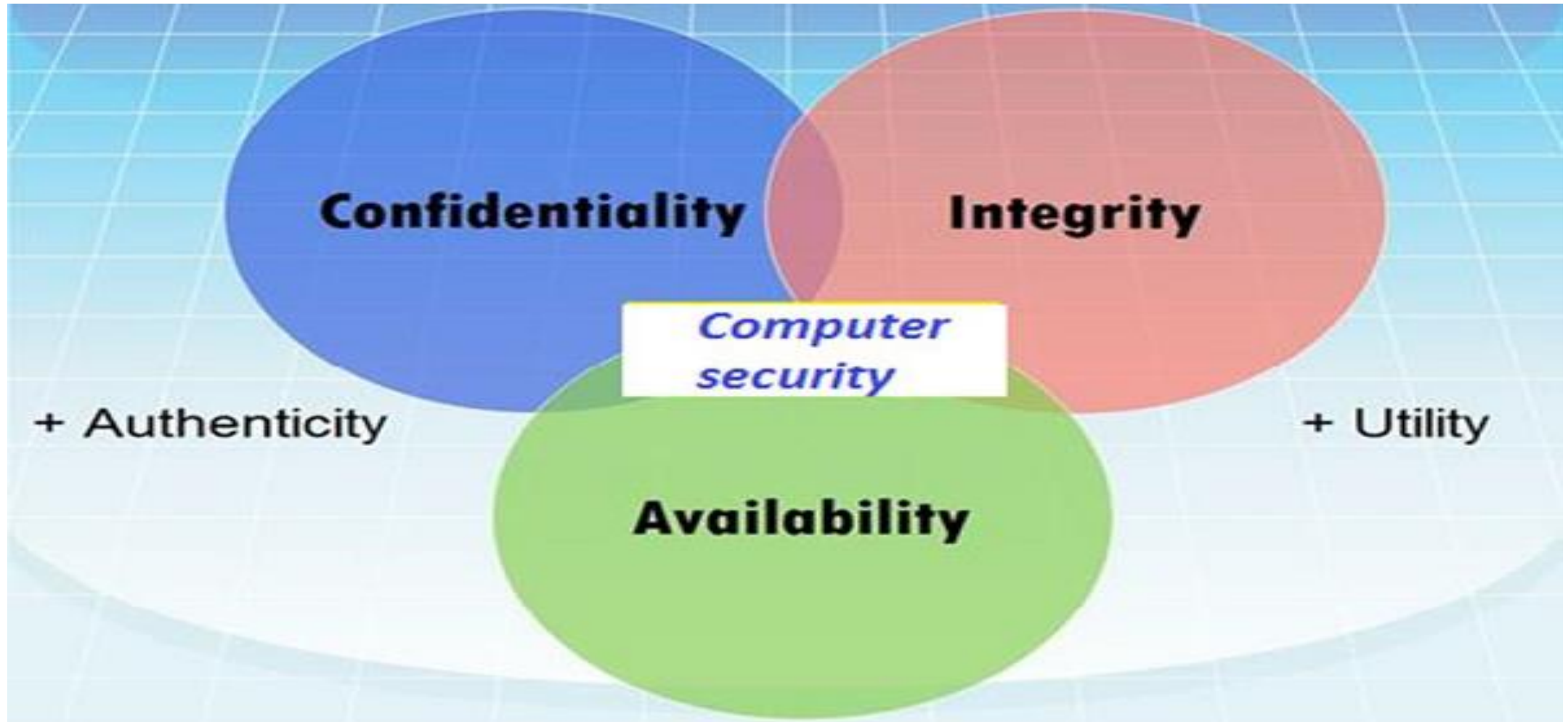
Reputation loss –

- Just think if your Facebook account or business email has been owned by a social engineering attack and it sends fake information to your friends, business partners. You will need time to gain back your reputation.

Identity theft –

- This is a case where your identity is stolen (photo, name surname, address, and credit card) and can be used for a crime like making false identity documents.

Different Elements in Computer Security



Different Elements in Computer Security cont..

- **Integrity**

Integrity is the trustworthiness of data in the systems or resources by the point of view of preventing unauthorized and improper changes.

- Generally, Integrity is composed of two sub-elements – data-integrity, which it has to do with the content of the data and authentication which has to do with the origin of the data as such information has values only if it is correct.

- **Availability**

- Availability refers to the ability to access data of a resource when it is needed, as such the information has value only if the authorized people can access at right time.
- Denying access to data nowadays has become a common attack. Imagine a downtime of a live server how costly it can be.

Different terminology used in Computer Security

- **Unauthorized access** – An unauthorized access is when someone gains access to a server, website, or other sensitive data using someone else's account details.
- **Hacker** – Is a Person who tries and exploits a computer system for a reason which can be money, a social cause, fun etc.
- **Threat** – Is an action or event that might compromise the security.
- **Vulnerability** – It is a weakness, a design problem or implementation error in a system that can lead to an unexpected and undesirable event regarding security system.
- **Attack** – Is an assault on the system security that is delivered by a person or a machine to a system. It violates security.
- **Antivirus or Antimalware** – Is a software that operates on different OS which is used to prevent from malicious software.
- **Social Engineering** – Is a technique that a hacker uses to stole data by a person for different for purposes by psychological manipulation combined with social scenes.
- **Virus** – It is a malicious software that installs on your computer without your consent for a bad purpose.
- **Firewall** – It is a software or hardware which is used to filter network traffic based on rules.

Security Mechanism

- A security mechanism is a method, tool, or procedure for enforcing a security policy.
- Mechanisms can be nontechnical, such as requiring proof of identity before changing a password; in fact, policies often require some procedural mechanisms that technology cannot enforce.

Security Mechanisms	
1. Encipherment	5. Traffic Padding
2. Digital Signature	6. Routing Control
3. Access Control	7. Notarization
4. Data Integrity	8. Authentication Exchange
06 Network Security	

Types of Security Mechanism

- **Encipherment :**

This security mechanism deals with hiding and covering of data which helps data to become confidential.

- It is achieved by applying mathematical calculations or algorithms which reconstruct information into not readable form.

- **Access Control :**

This mechanism is used to stop unattended access to data which you are sending. It can be achieved by various techniques such as applying passwords, using firewall, or just by adding PIN to data.

- **Notarization :**

This security mechanism involves use of trusted third party in communication.

- It acts as mediator between sender and receiver so that if any chance of conflict is reduced.
- This mediator keeps record of requests made by sender to receiver for later denied.

Security Mechanism Cont..

- **Data Integrity :**

This security mechanism is used by appending value to data to which is created by data itself.

- When this packet or data which is appended is checked and is the same while sending and receiving data integrity is maintained.

- **Authentication exchange :**

This security mechanism deals with identity to be known in communication.

- This is achieved at the TCP/IP layer where two-way handshaking mechanism is used to ensure data is sent or not

- **Bit stuffing :**

This security mechanism is used to add some extra bits into data which is being transmitted. It helps data to be checked at the receiving end and is achieved by Even parity or Odd Parity.

- **Digital Signature :**

This security mechanism is achieved by adding digital data that is not visible to eyes.

- It is form of electronic signature which is added by sender which is checked by receiver electronically.
- This mechanism is used to preserve data which is not more confidential but sender's identity is to be notified.

Cyber Law

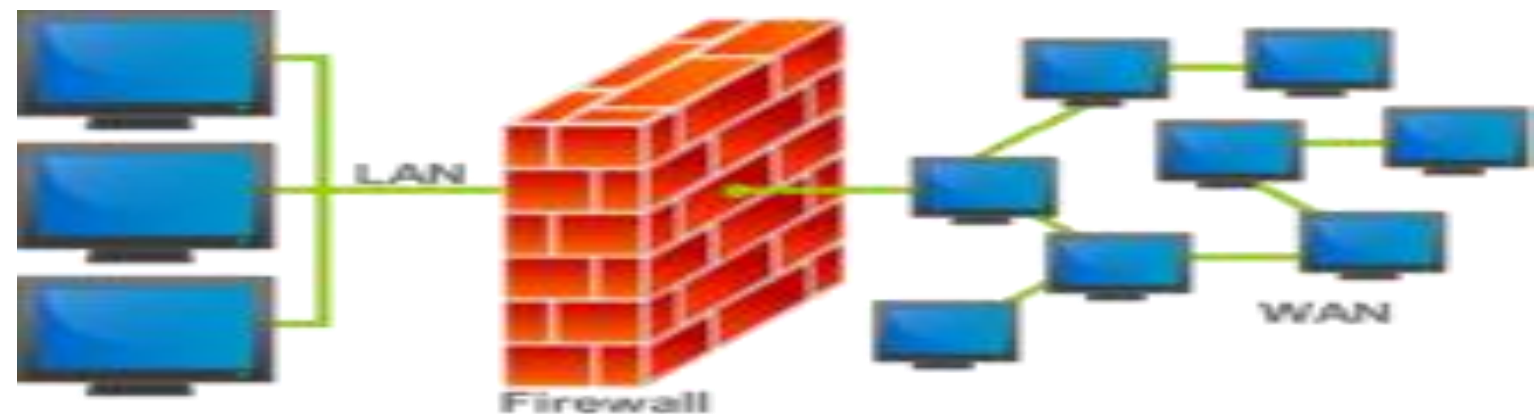
- **Cyber law** is the part of the overall legal system that deals with the Internet, cyberspace, and their respective legal issues.
- Cyber law covers a fairly broad area, encompassing several subtopics including freedom of expression, access to and usage of the Internet, and online privacy.
- Generically, cyber law is referred to as the Law of the Internet.
- It refers to all legal and regulatory aspects of internet and computers
- Primary purpose is to provide legal recognition of computer system and electronic act of a user
- In general, It is a law that bounds all the cyber crime.

Network Security

- Network security is a broad term that covers a multitude of technologies, devices and processes.
- In its simplest term, it is a set of rules and configurations designed to protect the integrity, confidentiality and accessibility of computer networks and data using both software and hardware technologies.
- Network security involves the authorization of access to data in a network.
- Network security covers a variety of computer networks, both public and private that are used in transaction of data.
- Network security helps to minimize cyber crime and data breaches

Firewall

- A Firewall is a network security device that monitors and filters incoming and outgoing network traffic based on an organization's previously established security policies. At its most basic, a firewall is essentially the barrier that sits between a private internal network and the public Internet.
- A firewall is a network security device that monitors incoming and outgoing network traffic and decides whether to allow or block specific traffic based on a defined set of security rules
- Fire wall can be a device or a computer program. It provides secure connectivity between networks
- Fire wall helps to block the unauthorized access and spams in a network.It helps to enhance the network security



How a Firewall Works



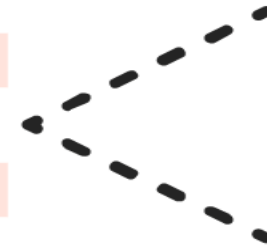
1

Your network
makes a request
to the internet



2

The firewall
scans and filters
the data



3

Your network is
granted access to the
requested source

Types of Firewalls



Next-generation
Firewalls

Web
Application
Firewalls

Network
Segmentation
Firewalls

Database
Firewalls

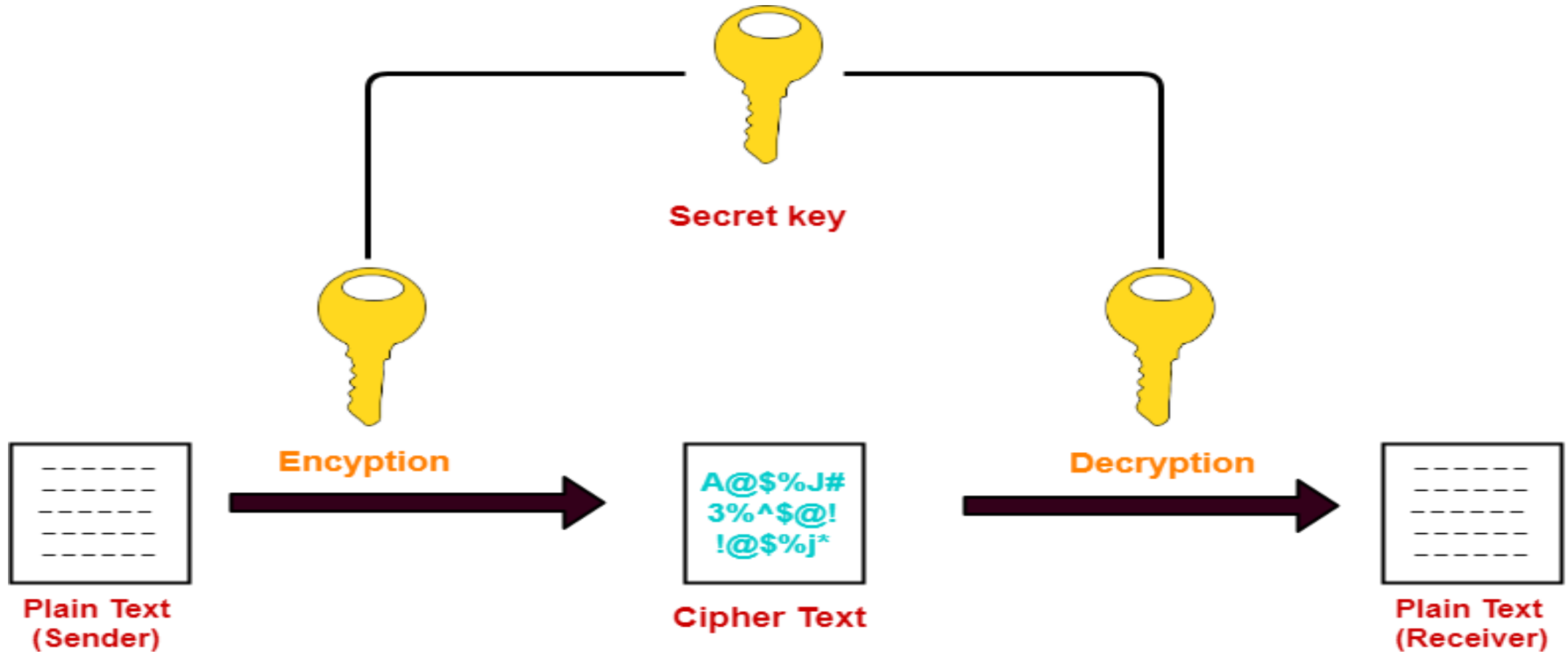
Cloud-based
Firewalls

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Cryptography

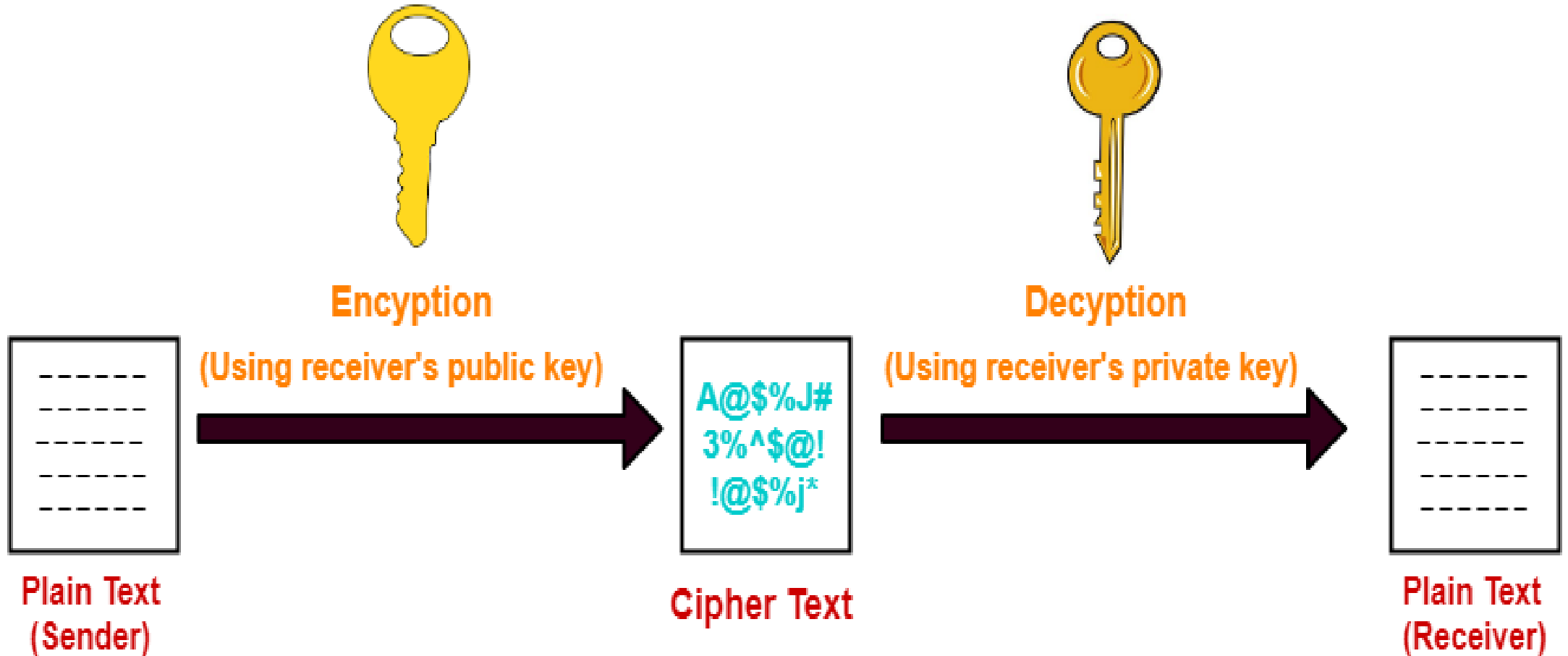
- Cryptography is the study of secure communications techniques that allow only the sender and intended recipient of a message to view its contents
- Cryptography is a method of protecting information and communications through the use of codes, so that only those for whom the information is intended can read and process it
- Cryptography is the study of securing communications from outside observers.
- Encryption algorithms take the original message, or plaintext, and converts it into ciphertext, which is not understandable.
- The key allows the user to decrypt the message, thus ensuring on they can read the message
- The techniques that cryptographers utilize can ensure the confidential transfer of private data.
- Techniques relating to digital signatures can prevent imposters from intercepting corporate data, while companies can use hash function techniques to maintain the integrity of data.

Cryptography Cont..



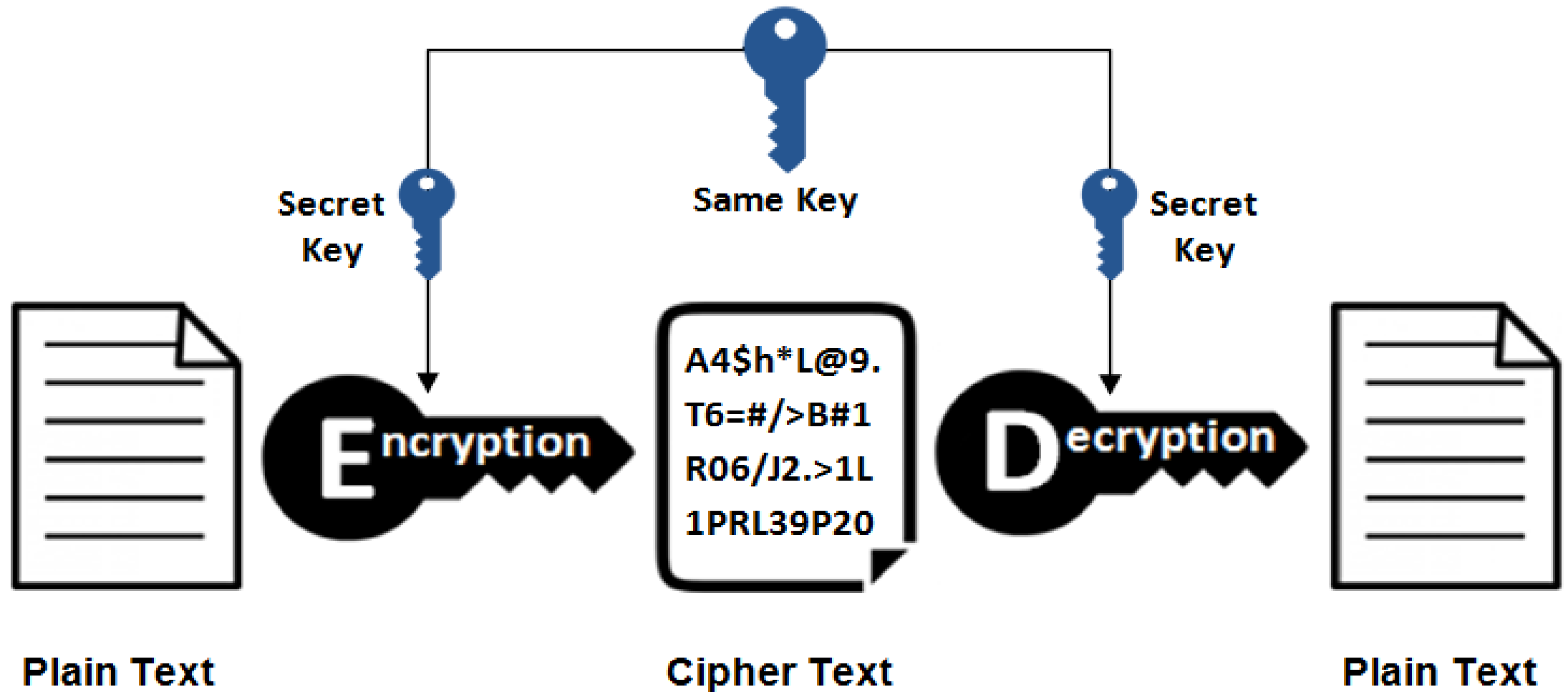
Symmetric Key Cryptography

Cryptography Cont..

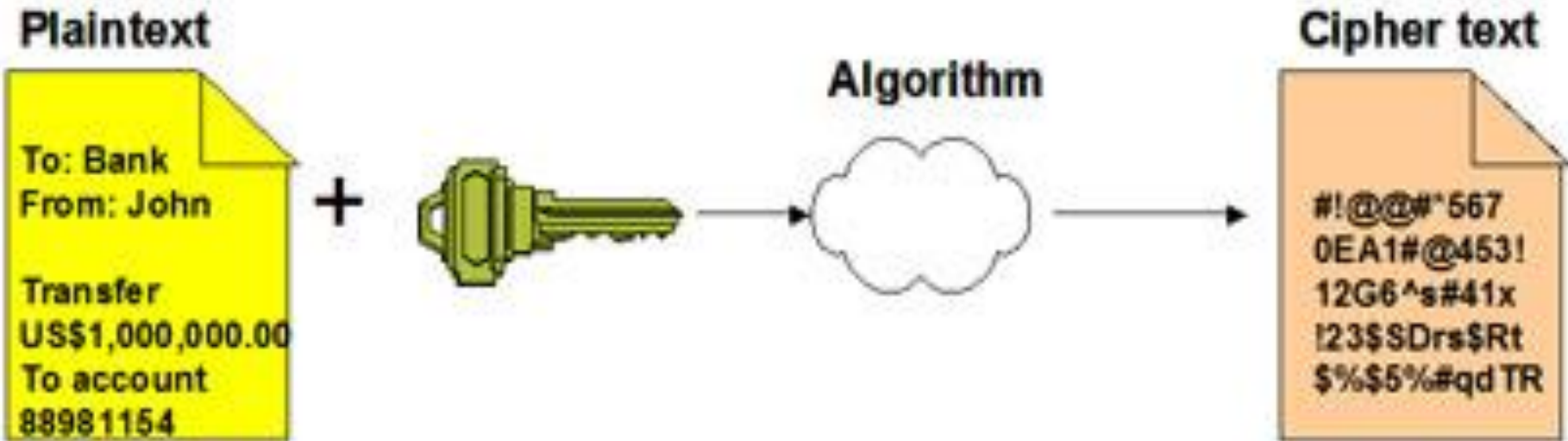


Asymmetric Key Cryptography

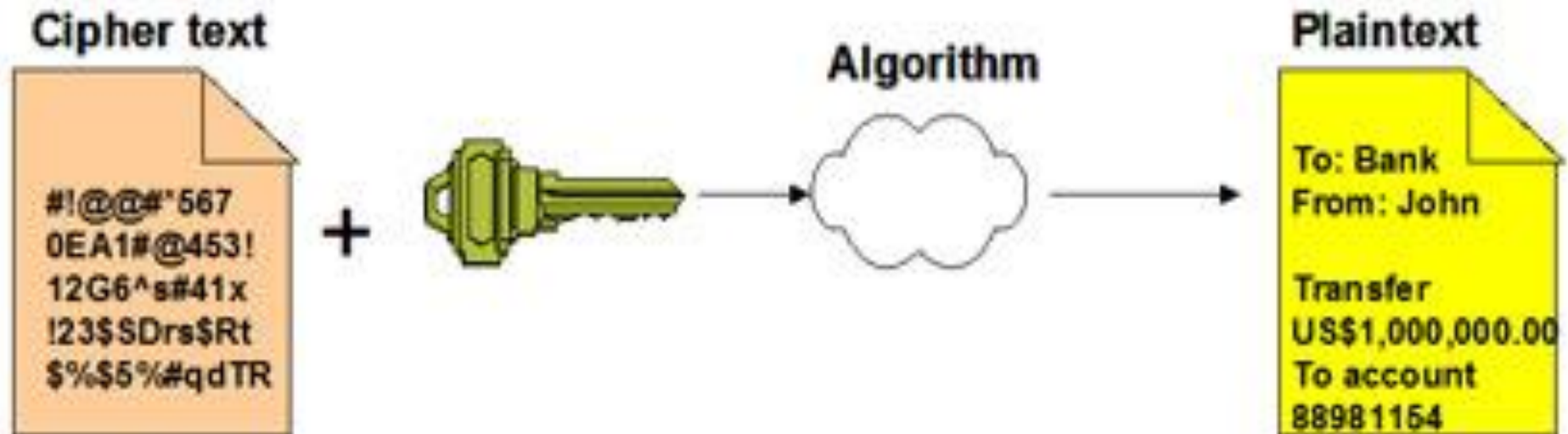
Symmetric Encryption



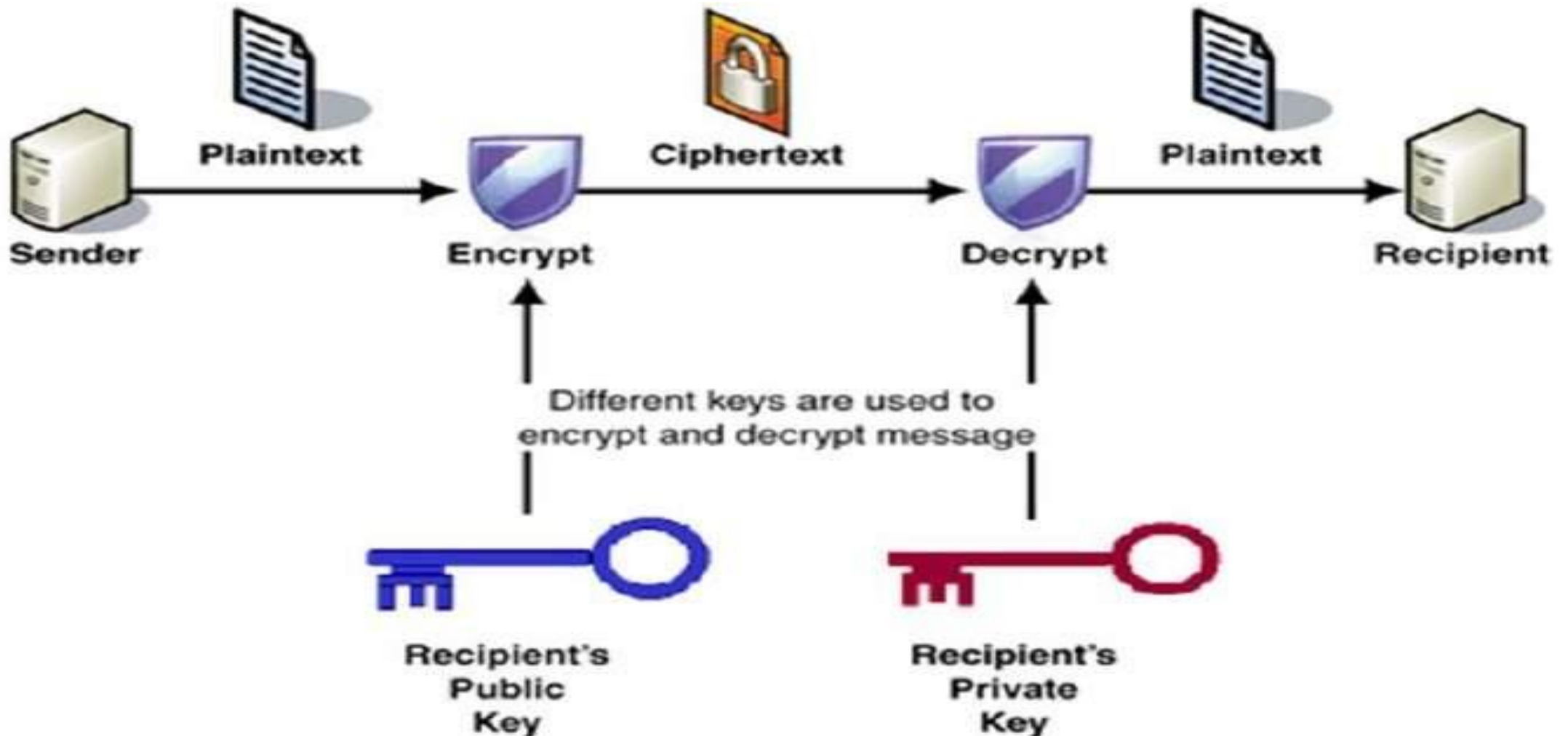
Encryption



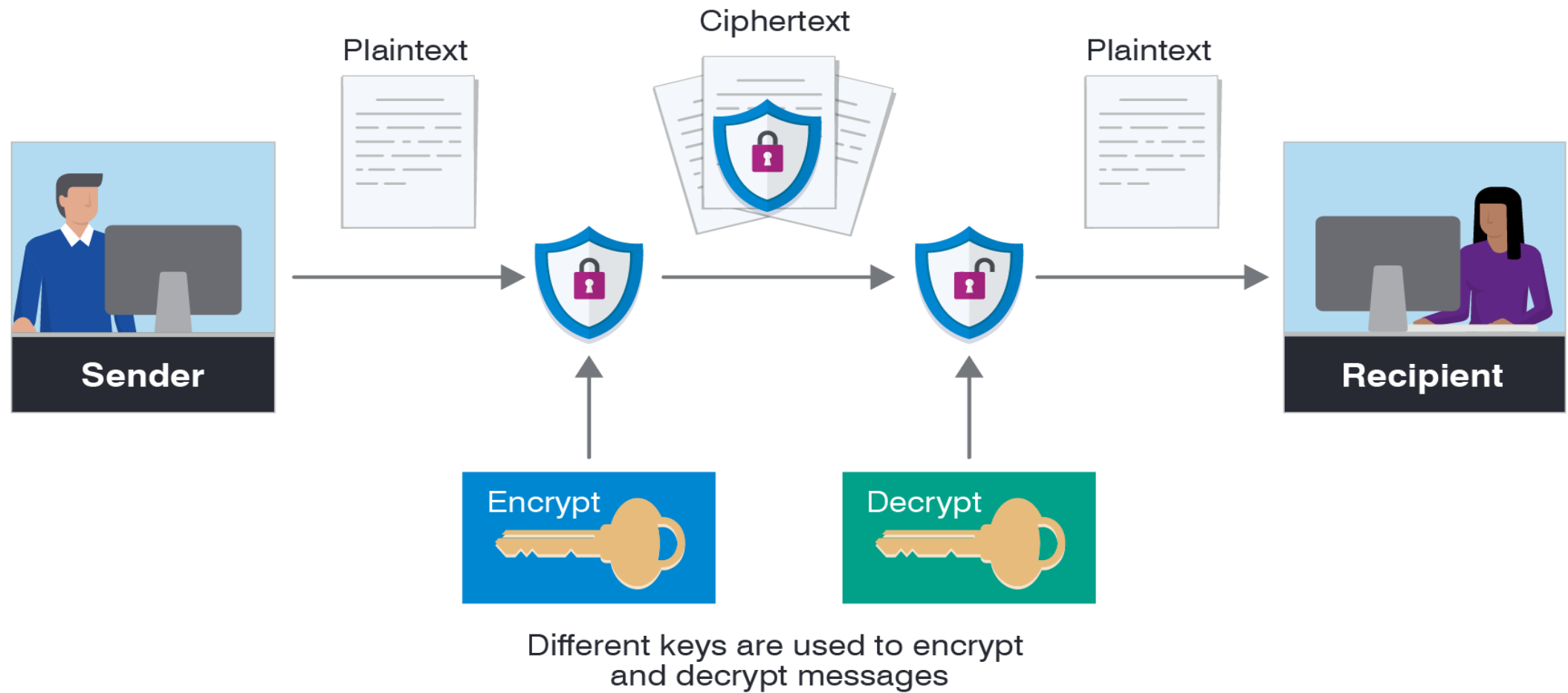
Decryption



Encryption and Decryption with a Asymmetric Key



Sample Encryption and Decryption Operation



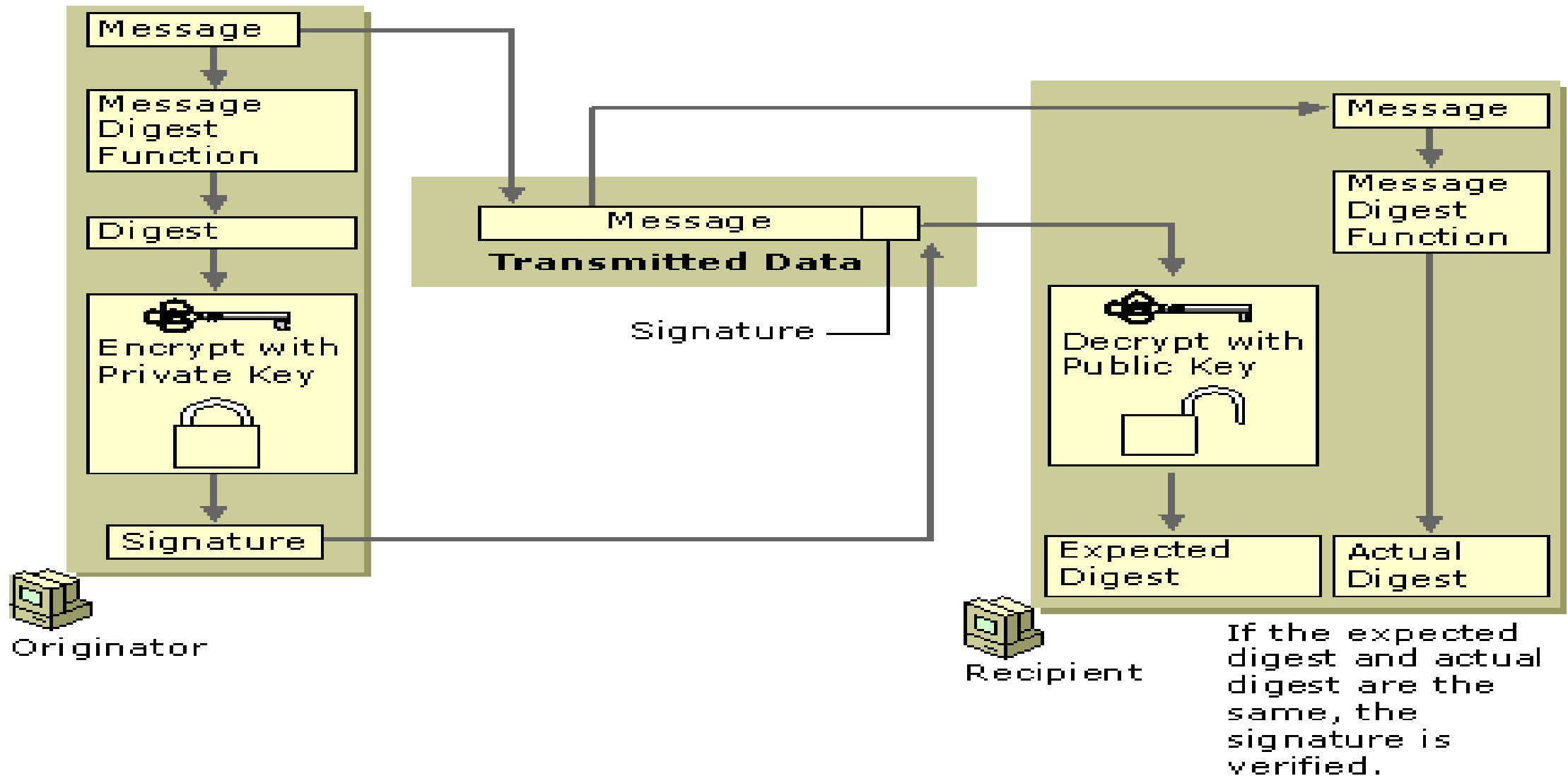
Digital Signature

- A digital signature uniquely identifies the originator of digitally signed data and also ensures the integrity of the signed data against tampering or corruption.
- One possible method for creating a digital signature is for the originator of data to create the signature by encrypting all of the data with the originator's private key and enclosing the signature with the original data.
- **However, encrypting all data to provide a digital signature is impractical for following two reasons:**
 - *The cipher text signature is the same size as the corresponding plaintext, so message sizes are doubled, consuming large amounts of bandwidth and storage space.*
 - *Public key encryption is slow and places heavy computational loads on computer processors*

Digital Signature Cont..

- Digital signature algorithms use more efficient methods to create digital signatures.
- The most common types of digital signatures today are created by signing **message digests** with the originator's private key to create a digital thumbprint of the data.
- Two of the most widely used digital signature algorithms today are the **RSA digital signature** process and the **Digital Signature Algorithm (DSA)**.
- **RSA Data Security Digital Signature Process:** In the RSA digital signature process, the private key is used to encrypt only the message digest.
- The encrypted message digest becomes the digital signature and is attached to the original data.

Basic RSA Data Security Digital Signature Process



Advantages of Digital Signature

- Higher security
- Legal compliance and wide acceptance
- Time savings
- Workflow automation
- Cost savings
- Happier end-users
- Better company image and CSR

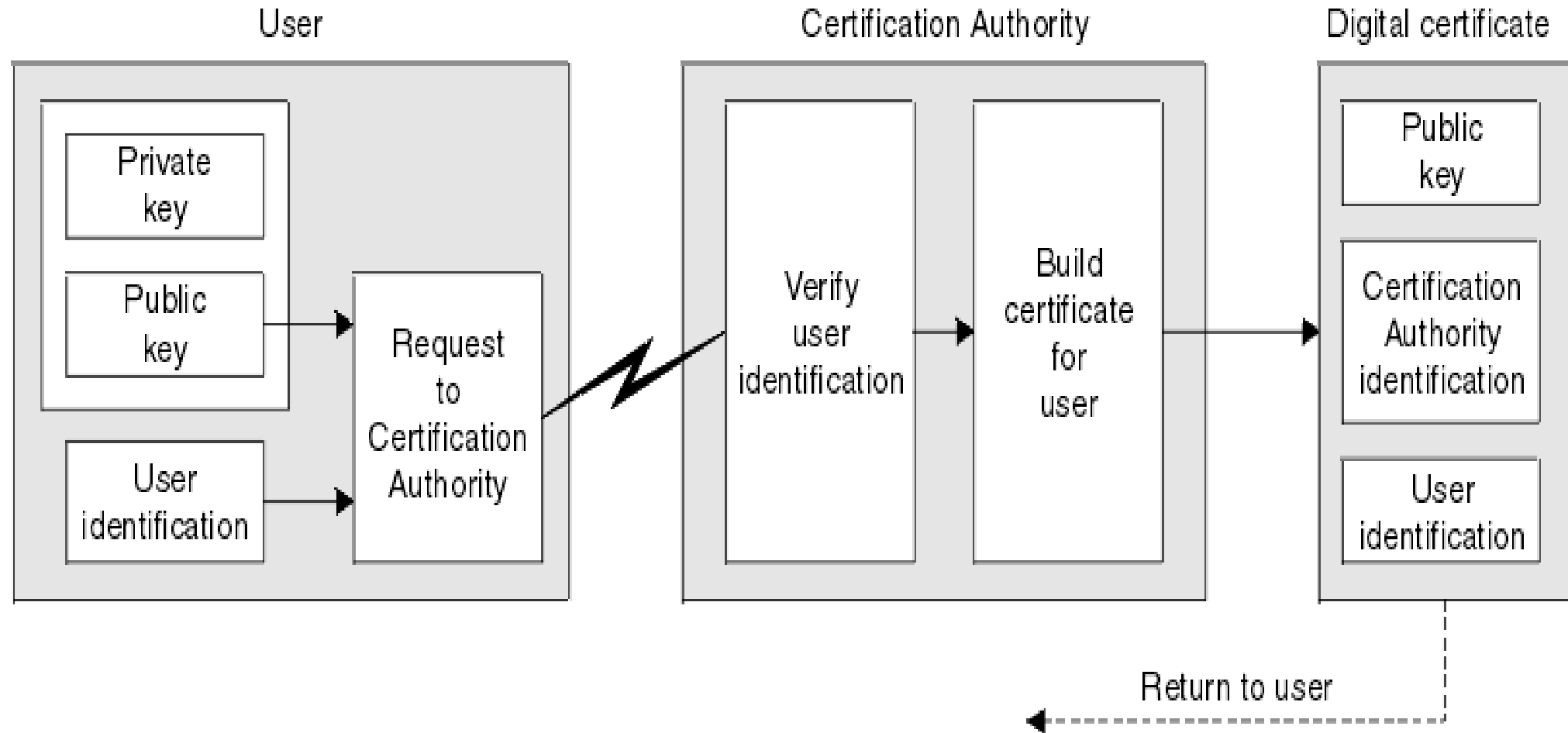
Digital Certificate and Certification Authority

- Digital certificates are electronic credentials that are used to assert the online identities of individuals, computers, and other entities on a network.
- **Digital certificates** function similarly to identification cards such as passports and drivers licenses.
- **Typically, certificates contain the following information:**
 - The subject's public key value
 - The subject's identifier information, such as the name and email address
 - The validity period (the length of time that the certificate is considered valid)
 - Issuer identifier information
 - The digital signature of the issuer, which attests to the validity of the binding between the subject's public key and the subject's identifier information

Digital Certificate and Certification Authority Cont..

- **Process to obtain a Certificate From CA:**
- User can generate a Key pair of its own and generate a Certificate Signing Request (CSR) and then send the CSR to Issuing CA for a certificate.
- CSR contains the public key of the user and user identity information in a format that issuing CAs would normally expect as shown in figure below.

Digital Certificate and Certification Authority Cont..



Digital Certificate and Certification Authority

- A **Certificate Authority (CA)** issues digital certificates that contain a public key and the identity of the owner.
- The matching private key is not made available publicly, but kept secret by the end user who generated the key pair.
- The certificate is also a confirmation or validation by the CA that the public key contained in the certificate belongs to the person, organization, server or other entity noted in the certificate.
- In essence, the Certificate Authority is responsible for saying "yes, this person is who they say they are, and we, the CA, verify that".

Digital Certificate and Certification Authority Cont..

- **Third Party Authentication**

- In third-party authentication systems, the password or encryption key itself never travels over the network.
- Rather, an "authentication server" maintains a file of obscure facts about each registered user.
- There are many variations on this theme.
- **Kerberos:** Kerberos is a popular third-party authentication protocol.
- Kerberos is an encryption-based system that uses secret key encryption designed to authenticate users and network connections

The authentication process

- Client A sends a request to the Kerberos authentication server (KAS) requesting "credentials" for a given server, B.
- **The KAS responds with the following information, which is encrypted in A's key:**
 - A "ticket" for the server. This ticket contains B's key.
 - A temporary encryption key (often called a "session key").
 - A then transmits—the client's identity and a copy of the session key, both encrypted in B's key—to B.
 - The session key (now shared by the client and server) is used to authenticate the client and used to authenticate the server in future transaction.
 - The session key is then used to encrypt further communication between the two parties or to exchange a separate sub-session key to be used to encrypt further communication.

Security Awareness

- Security awareness is the knowledge and attitude members of an organization possess regarding the protection of the physical, and especially informational, assets of that organization
- Security awareness training is a strategy used **by IT and security professionals to prevent and mitigate user risk.**
- A security awareness program is **a formal program with the goal of training users of the potential threats to an organization's information and how to avoid situations** that might put the organization's data at risk.

The key elements of a security awareness program

- Research.
- Development.
- Implementation.
- Monitoring and Review.

Security Policy

- Security policy is a definition of what it means to be secure for a system, organization or other entity.
- For an organization, it addresses the constraints on behavior of its members as well as constraints imposed on adversaries by mechanisms such as doors, locks, keys and wall
- A security policy is of no use to an organization or the individuals within an organization if they cannot implement the guidelines or regulations within the policy.
- It should be **concise, clearly written and as detailed as possible** in order to provide the information necessary to implement the regulation
- There are 2 types of security policies:
- **technical security and administrative security policies.**
- Technical security policies describe the configuration of the technology for convenient use; body security policies address however all persons should behave. All workers should conform to and sign each the policies.

Security Policy Cont..

- An IT Security Policy **identifies the rules and procedures for all individuals accessing and using an organization's IT assets and resources.**
- Effective IT Security Policy is a model of the organization's culture, in which rules and procedures are driven from its employees' approach to their information and work.

Five Components of Security Policy:

- It relies on five major elements: **confidentiality, integrity, availability, authenticity, and non-repudiation.**



Thank you!